



Solving Quadratic Equations by Factoring

Algebra Worksheet · Grade 8-10

Name: _____

Date: _____

Score: / 10

Learning Objectives

- Factor quadratic expressions of the form ax^2+bx+c
- Apply the Zero Product Property to find solutions
- Verify solutions by substitution into the original equation

Solve each quadratic equation by factoring and show all work on a separate sheet.

1. Solve the quadratic equation by factoring.

$$x^2 + 5x + 6 = 0$$

Answer: _____

2. Solve the quadratic equation by factoring.

$$x^2 - 7x + 12 = 0$$

Answer: _____

3. Solve the quadratic equation by factoring.

$$x^2 - 9 = 0$$

Answer: _____

4. Solve the quadratic equation by factoring.

$$2x^2 + 7x + 3 = 0$$

Answer: _____

5. Solve the quadratic equation by factoring.

$$x^2 + 4x = 0$$

Answer: _____

6. Solve the quadratic equation by factoring.

$$3x^2 - 12x + 9 = 0$$

Answer: _____

7. Solve the quadratic equation by factoring.

$$x^2 - 10x + 25 = 0$$

Answer: _____



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Labanan ang agwat.

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8. Solve the quadratic equation by factoring.

$$6x^2 - x - 2 = 0$$

Answer: _____

9. Solve the quadratic equation by factoring.

$$x^2 + 2x - 15 = 0$$

Answer: _____

10. Solve the quadratic equation by factoring.

$$4x^2 - 25 = 0$$

Answer: _____





Encourage students to always set the equation equal to zero before attempting to factor.

Solutions

1. Solve the quadratic equation by factoring.

$$x^2 + 5x + 6 = 0$$

→ Find two numbers that multiply to 6 and add to 5, which are 2 and 3.

→ Factor the expression as $(x+2)(x+3)=0$.

→ Set each factor equal to zero to get $x=-2$ or $x=-3$.

Answer: $x = -2, x = -3$

2. Solve the quadratic equation by factoring.

$$x^2 - 7x + 12 = 0$$

→ Find two numbers that multiply to 12 and add to -7, which are -3 and -4.

→ Factor the expression as $(x-3)(x-4)=0$.

→ Set each factor equal to zero to get $x=3$ or $x=4$.

Answer: $x = 3, x = 4$

3. Solve the quadratic equation by factoring.

$$x^2 - 9 = 0$$

→ Recognize the difference of squares pattern.

→ Factor as $(x-3)(x+3)=0$.

→ Set each factor equal to zero to get $x=3$ or $x=-3$.

Answer: $x = 3, x = -3$

4. Solve the quadratic equation by factoring.

$$2x^2 + 7x + 3 = 0$$

→ Multiply a and c to get 6, then find two numbers that multiply to 6 and add to 7, which are 1 and 6.

→ Rewrite the middle term and factor by grouping to get $(2x+1)(x+3)=0$.

→ Set each factor equal to zero to get $x=-1/2$ or $x=-3$.

Answer: $x = -\frac{1}{2}, x = -3$

5. Solve the quadratic equation by factoring.

$$x^2 + 4x = 0$$

→ Factor out the common factor x to get $x(x+4)=0$.

→ Set each factor equal to zero.

→ Solve to find $x=0$ or $x=-4$.

Answer: $x = 0, x = -4$



6. Solve the quadratic equation by factoring.

$$3x^2 - 12x + 9 = 0$$

→ Factor out the greatest common factor 3 to get $3(x^2 - 4x + 3) = 0$.

→ Factor the trinomial as $3(x-1)(x-3) = 0$.

→ Set each factor equal to zero to get $x=1$ or $x=3$.

Answer: $x = 1, x = 3$

7. Solve the quadratic equation by factoring.

$$x^2 - 10x + 25 = 0$$

→ Recognize the perfect square trinomial pattern.

→ Factor as $(x-5)^2 = 0$.

→ Solve to find the double root $x=5$.

Answer: $x = 5$

8. Solve the quadratic equation by factoring.

$$6x^2 - x - 2 = 0$$

→ Multiply a and c to get -12, then find two numbers that multiply to -12 and add to -1, which are -4 and 3.

→ Rewrite the middle term and factor by grouping to get $(3x-2)(2x+1) = 0$.

→ Set each factor equal to zero to get $x=2/3$ or $x=-1/2$.

Answer: $x = \frac{2}{3}, x = -\frac{1}{2}$

9. Solve the quadratic equation by factoring.

$$x^2 + 2x - 15 = 0$$

→ Find two numbers that multiply to -15 and add to 2, which are 5 and -3.

→ Factor the expression as $(x+5)(x-3) = 0$.

→ Set each factor equal to zero to get $x=-5$ or $x=3$.

Answer: $x = 3, x = -5$

10. Solve the quadratic equation by factoring.

$$4x^2 - 25 = 0$$

→ Recognize the difference of squares pattern with $(2x)^2$ and 5^2 .

→ Factor as $(2x-5)(2x+5) = 0$.

→ Set each factor equal to zero to get $x=5/2$ or $x=-5/2$.

Answer: $x = \frac{5}{2}, x = -\frac{5}{2}$

